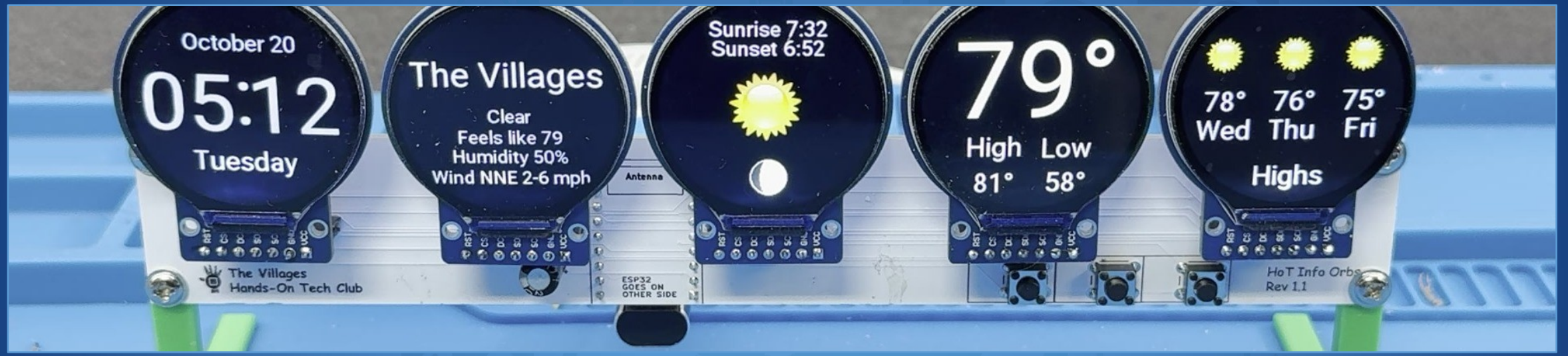
# HOT Info Orbs

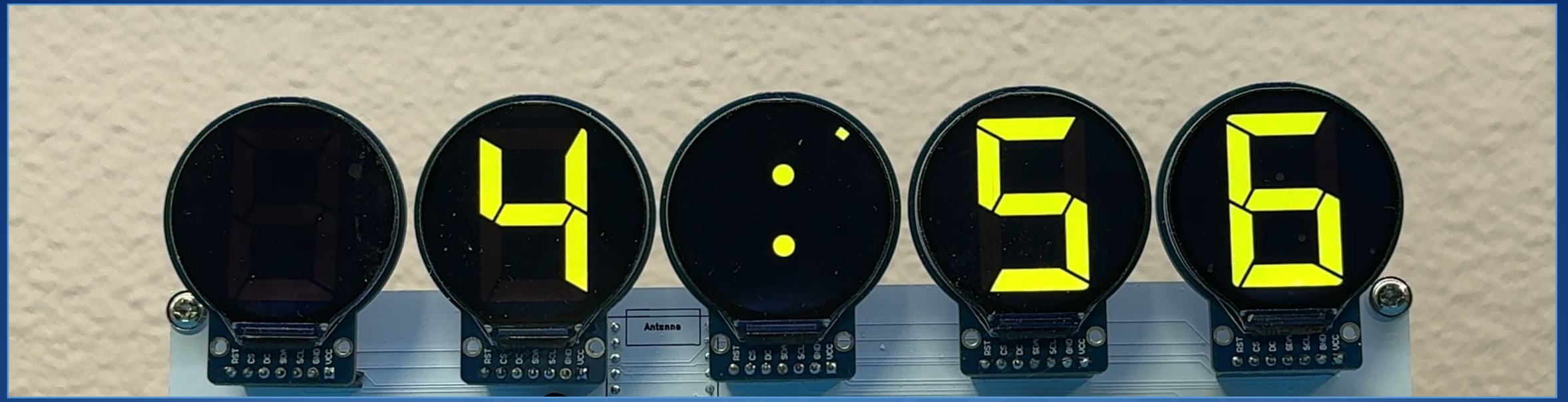
Five round screens that show what you want to know




**Hands-On Tech, Small Computers group** · October 27, 2026 · Scott Dube

A club build: a board designed by Yang and Scott, firmware and docs on GitHub

# What it is




**Five round screens**

Color displays an inch and a quarter across, on one board. Each page uses all five.

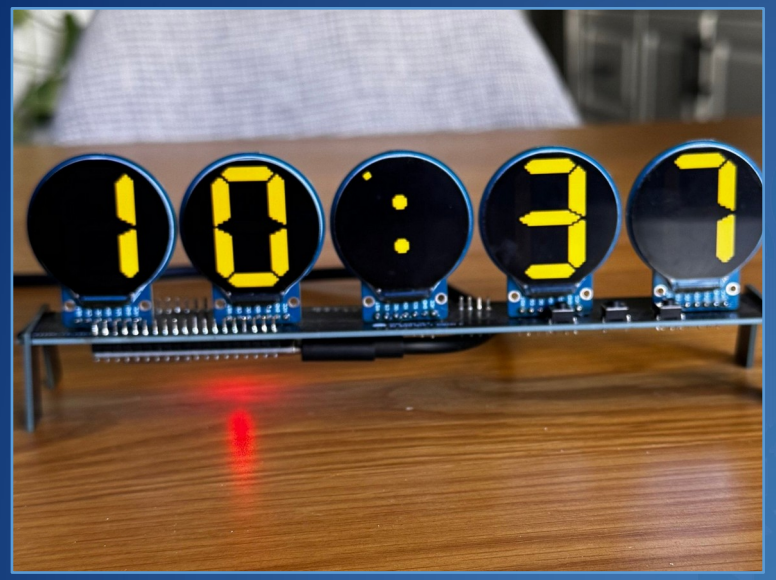
**One small computer**

A XIAO ESP32-S3 with Wi-Fi fetches the time, the weather, the markets and your own data.

**Nothing to install**

A USB cable for power, three buttons, and a settings page you open on your phone.

# Where it came from




## The original: Info Orbs

Brett Turner designed the Info Orbs and shared the plans and software for free.

Photo: Brett Turner, brett.tech




## The kit: HackerBox 0129 “Orbital”

HackerBoxes, a monthly electronics kit, sent out its own version.

Image: HackerBoxes




## Built in the club

A couple of our members built that kit. We liked it enough to make our own: a new board, new features, a build anyone can follow.

Photo: Donald Champion

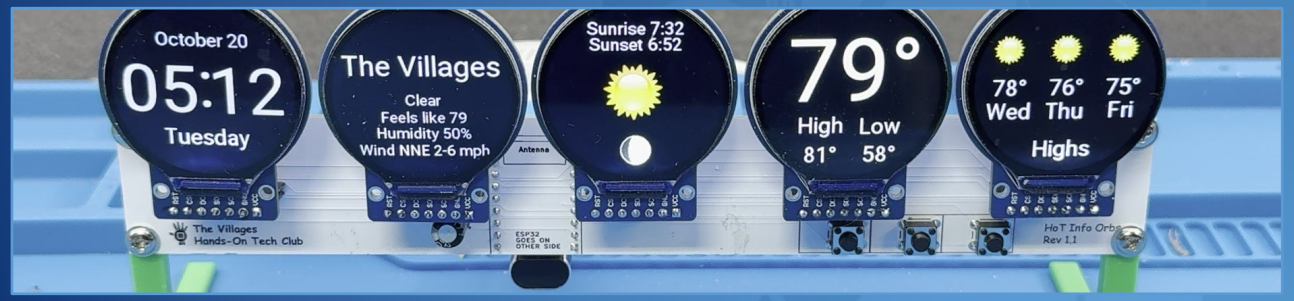
# The pages: clock and weather



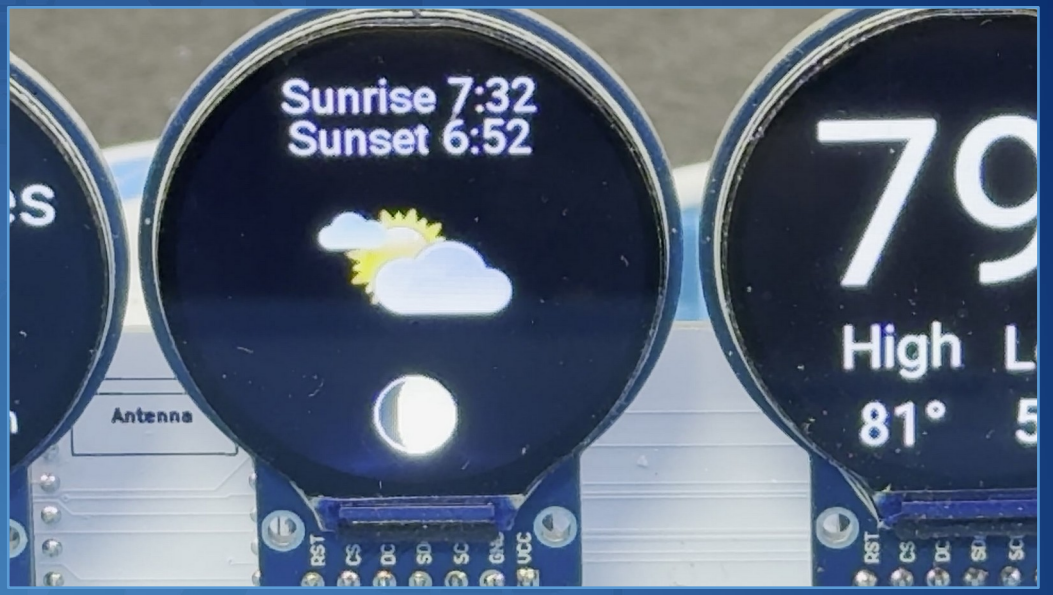

**Clock, LED digits**




**Clock, Nixie face (middle button)**




**Weather: now, today, and the next three days**




**Sunrise, sunset and tonight's moon**

The clock spreads across all five orbs: 12- or 24-hour, in any colors you like.

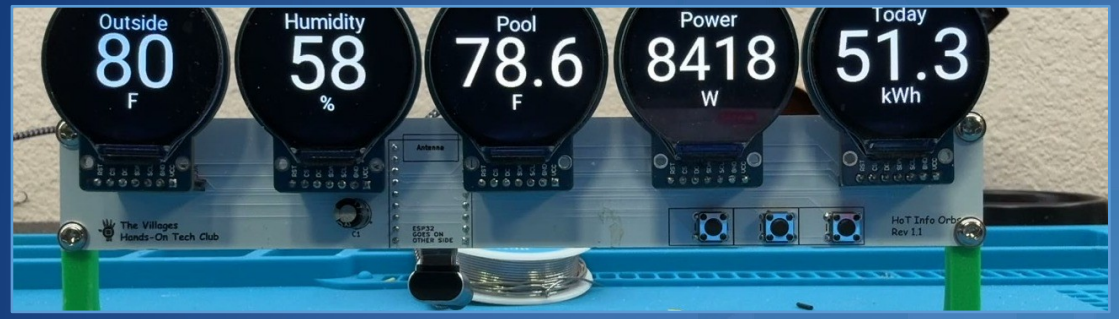
Weather for any town, or straight from your own Tempest station. Up to two stations.

# The pages: stocks and your own data




## Stocks

Five tickers, one per orb: price, change and the day's range. Stocks, funds or crypto.




## MQTT

Five live values from Home Assistant, or anything else that speaks MQTT.

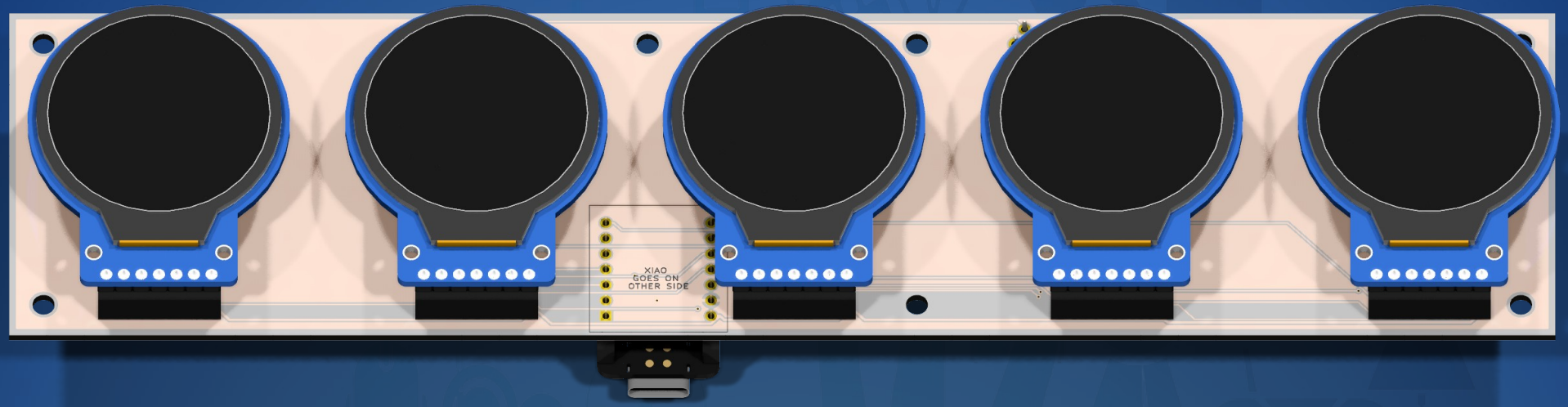



## Web Data

A small JSON file from any web address, laid out the way you want.

Left and right buttons step through the pages. Leave it alone and it rotates on its own. The last two pages are ones you build yourself.

# How it’s built




**XIAO ESP32-S3 or S3 Plus**
the brain, Wi-Fi built in, in a socket

**Five 1.28 in displays**
GC9A01 round color TFTs, socketed

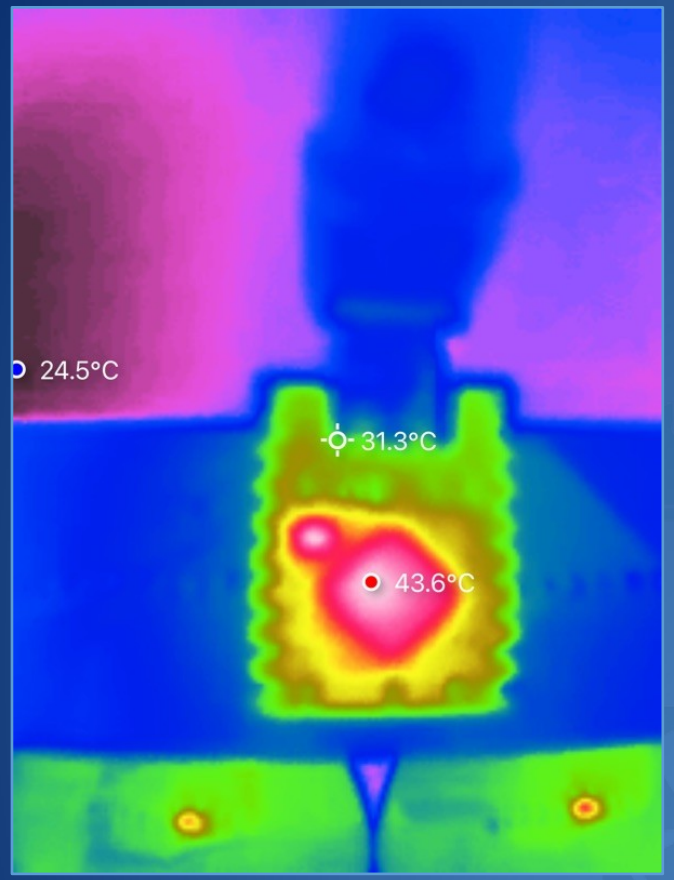
**Three buttons**
pages, clock face, wake

**USB-C**
power and the first firmware load

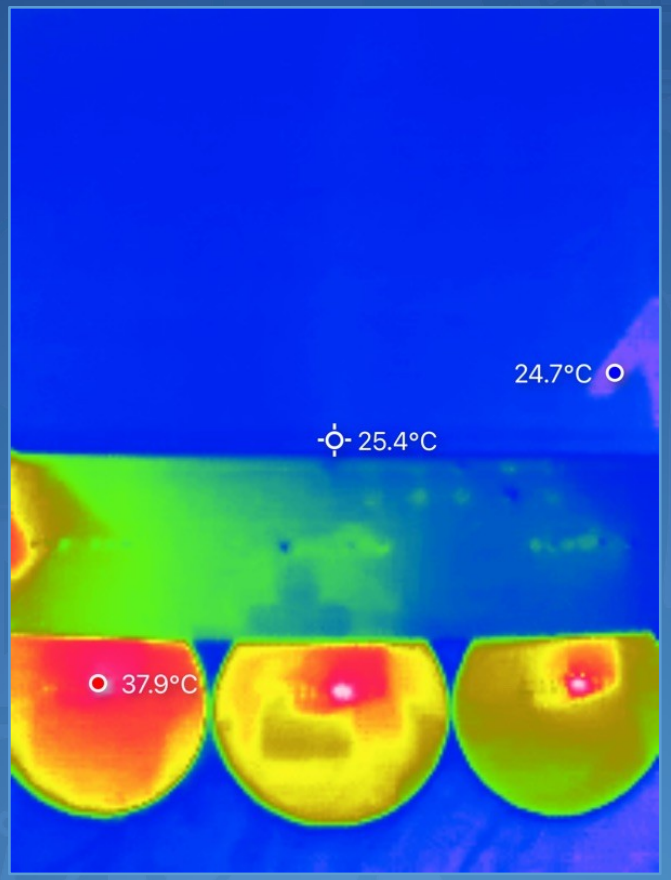
**Under 2 W**
1.7 W measured at 5 V

**Board by Yang and Scott** · made by JLCPCB · every file on GitHub

# Why the displays run on 5 V




The ESP32 module, hottest point




The backs of three displays

✔ **43.6 °C**

our module, after hours of running

✔ **37.9 °C**

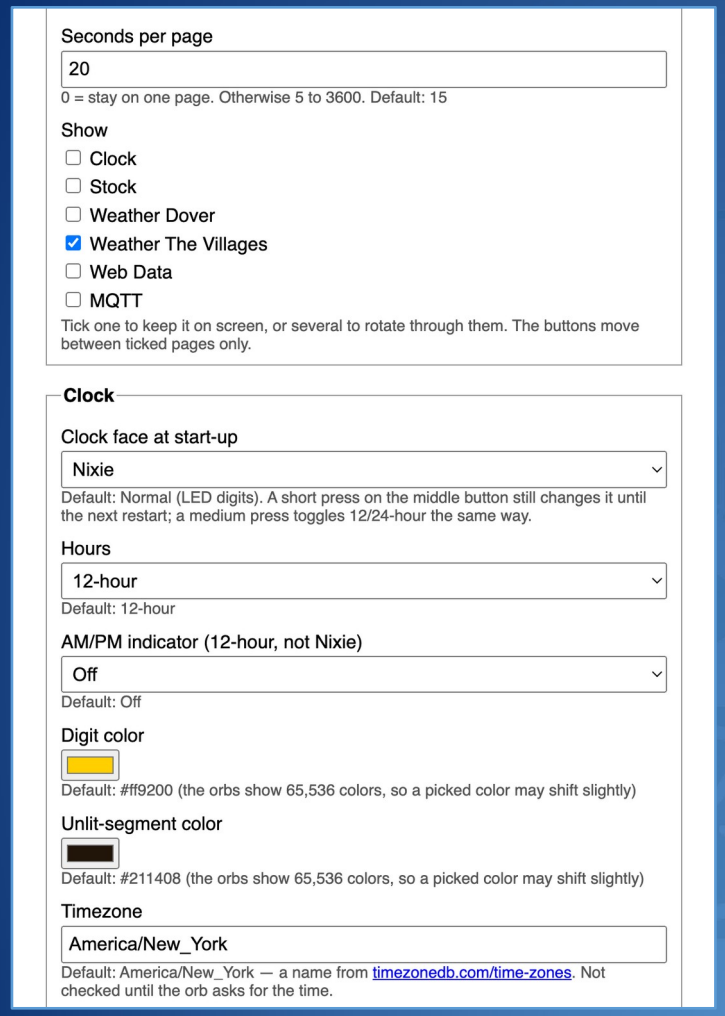
our display backs

✖ **82 °C**

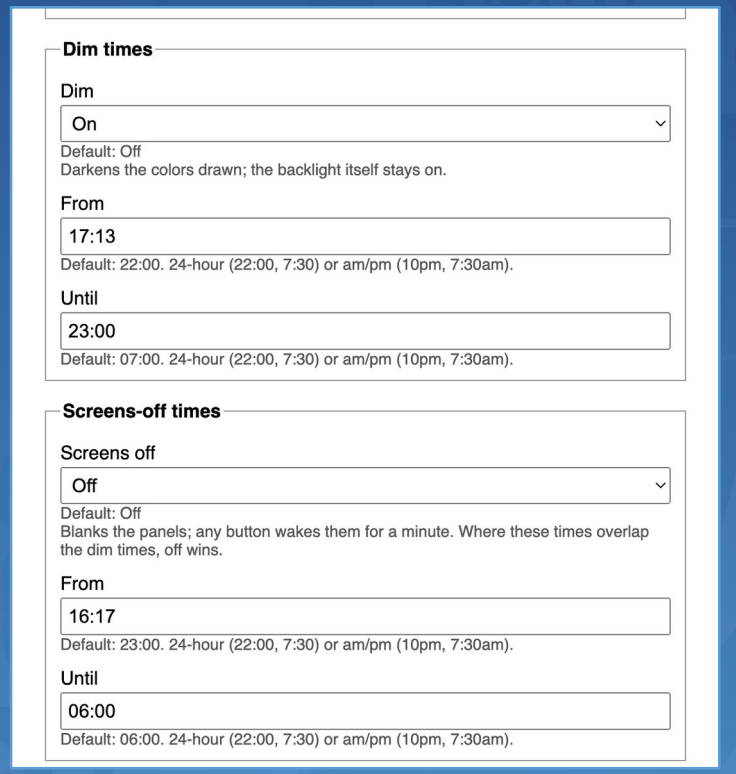
the kit’s module: “too hot to touch”, a builder’s report

On the kit every display ran off the ESP’s regulator. Here each display has its own 5 V feed, so they carry their own share and stay warm, not hot.

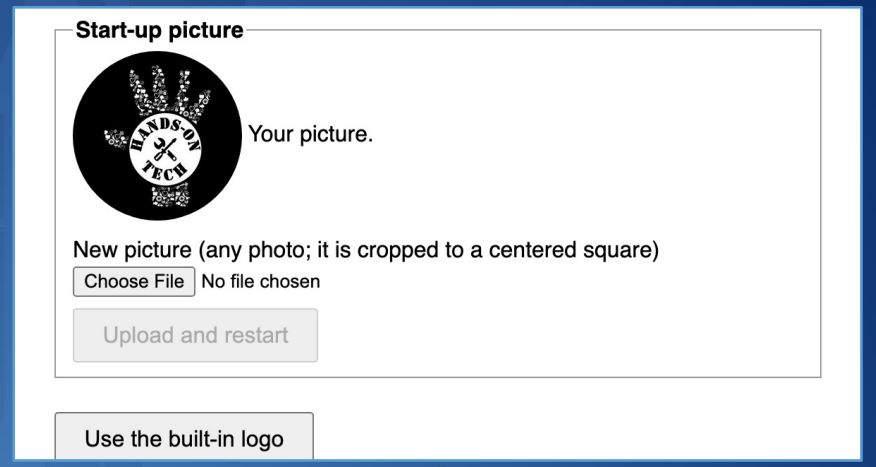
# Setting it up from your phone




Pages and clock



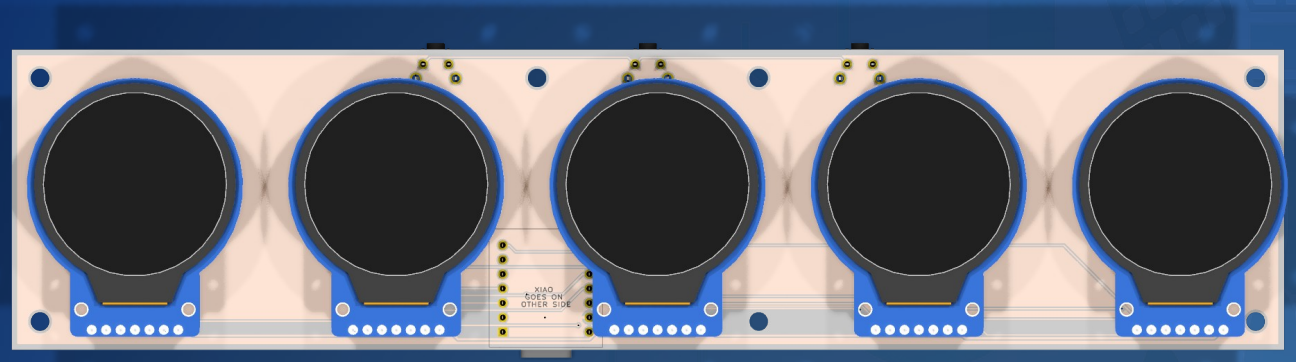

Dim and screens-off times




Your own start-up picture

- Tick the pages you want; untick the rest
- Your town, tickers, time zone, colors
- Dims in the evening, screens off at bedtime; any button wakes it
- Firmware loads once over USB, then updates over Wi-Fi from this page

# What’s next



## ✓ Board rev 2.4: design locked

XIAO ESP32-S3 or S3 Plus, 46 mm display pitch to match the enclosure, buttons on top. This is the board the club builds.




## ○ Enclosure: concept, by Yang

“Winged Speedster”: brass rings around each screen, finned end caps, all 3D printed. Still a design.

# The club build

## 1 Get the parts

Board, XIAO ESP32-S3 or S3 Plus, five displays, sockets, buttons, a USB-C cable.

## 2 Build night

Sockets and headers only: through-hole soldering, one evening.

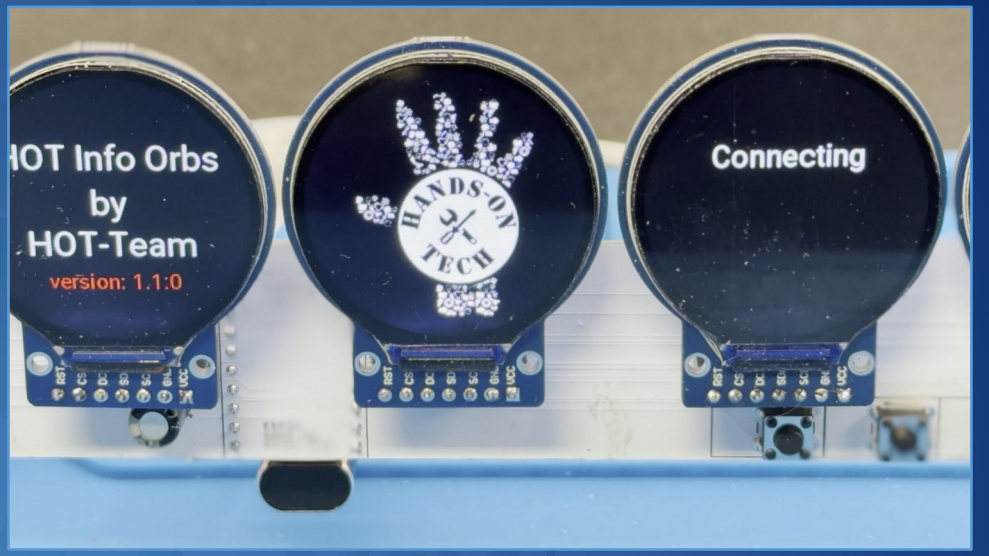
## 3 Load the firmware

Once, over USB. After that, updates from your phone.

## 4 Set it up

Pick your pages on the settings page. No code.

**Starting in January. Say tonight if you want in.**




**Boots with the club logo**

or any picture you upload. The module lifts out of its socket, so nothing is soldered that can’t be swapped.

# Links and credits

## Everything is on GitHub

github.com/scottdube/hot-info-orbs

Board files, firmware, build guide

## Feature sheet

scottdube.github.io/hot-info-orbs/features.html

## Thanks to

**Brett Turner** Info Orbs, the original

**HackerBoxes** #0129 Orbital

**Donald Champion** HackerBox build photos

**Yang and Scott** board design

**Yang** enclosure

**The members who built the first kit**

# Questions?

The orb is on the table. Press the buttons.




github.com/scottdube/hot-info-orbs